ONLINE SUPPLEMENTARY APPENDICES

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ABBREVIATIONS

AAs=amino acids, admin=administration, BCAAs=branched-chain amino acids, BID=twice per day, D5W=5% dextrose in water, F=follow-up, HCC=hepatocellular carcinoma, IQR=interquartile range, IV=intravenous, LCT=long-chain triglycerides, LOS=length of stay, MCTs=medium-chain triglycerides, mos=months, NA=not applicable, No.=number, NR=not reported, NS=normal saline, PN=parenteral nutrition, post-op=post-operative, QOL=quality of life, SD=standard deviation, SE=standard error, SF=Short-Form, TPN=total parenteral nutrition, tx=treatment, var=variance, yrs=years.

APPENDIX 1. MEDLINE Search Strategy (Ovid; 1946-present)

1. fish oils/ or fatty acids, omega-3/ or docosahexaenoic acids/ or eicosapentaenoic acid/
2. arginine/
3. glutamine/
4. RNA/
5. exp Mistletoe/ or Viscum album/
6. vitamin k/ or vitamin k 1/ or vitamin k 2/ or vitamin k 3/
7. probiotics/ or synbiotics/ or exp Lactobacillus/ or exp Bifidobacterium/ or Saccharomyces boulardii/
8. amino acids, branched-chain/ or isoleucine/ or leucine/ or valine/
9. Lycopersicon esculentum/
10. Ubiquinone/
11. ((fish adj oil*) or (omega adj 3*) or docosahexaenoic or eicosapentaenoic or immunonutrition or arginine or glutamine or mistletoe or Viscum album or (vitamin adj k*) or probiotic* or synbiotic* or Lactobacill* or Bifidobacteri* or Saccharomyces boulardii or branched chain amino acid* or leucine or isoleucine or valine or tomato or lycopene or coq10 or (coenzyme adj q10) or ubiquino*).ti,ab,kf.
12. or/1-11
13. (cancer* or carcinoma* or tumor* or tumour* or neoplas* or adenoma* or metastas* or malignan* or oncolog*).mp.
14. exp Neoplasms/
15. exp Precancerous Conditions/
16. 14 not 15
17. ((anti adj cancer) or anticancer). ti,ab,kf.
18. ((anti adj neoplastic) or antineoplastic). ti,ab,kf.
19. exp Antineoplastic Agents/
20. 13 or 16 or 17 or 18 or 19
21. 12 and 20
22. limit 12 to cancer
23. 21 or 22
24. exp perioperative care/ or exp perioperative period/ or exp Specialties, Surgical/ or exp Surgical Procedures, Operative/ or exp Postoperative Complications/
25. (surgery or surgical or surgeries or resect* or perioperative* or preoperative* or operative* or postoperative*). ti,ab,kf.
26. 24 or 25
27. 23 and 26
28. exp Animals/ not (exp Animals/ and Humans/)
29. 27 not 28
30. limit 29 to english language
31. (randomized controlled trial or controlled clinical trial).pt.
32. (randomized or randomly or placebo).ab.
33. (trial or groups).ab.
34. drug therapy.fs.

35. 31 or 32 or 33 or 34
36. 30 and 35
37. exp case-control studies/ or exp cohort studies/
38. longitudinal studies/ or follow-up studies/ or prospective studies/ or retrospective studies/ or Control Groups/ or Matched-Pair Analysis/ or Comparative studies/
39. 37 or 38
40. 30 and 39
41. limit 30 to (clinical trial, phase ii or clinical trial, phase iii or clinical trial, phase iv or comparative study or controlled clinical trial or observational study or randomized controlled trial)
42. 36 or 40 or 41

APPENDIX 2. Tables of Results from RCTs and Alphabetical List of Included RCTs

Note: The number below corresponds to the reference number cited in the main report's References list:

53. Beppu T, Beppu T, Nitta H, et al. Effect of branched-chain amino acid supplementation on functional liver regeneration in patients undergoing portal vein embolization and sequential hepatectomy: a randomized controlled trial. J Gastroent. 2015;50(12):1197-1205..

54. Bonau RA, Bonau RA. High-branched chain amino acid solutions: relationship of composition to efficacy. JPEN J Parenter Enteral Nutr. 1984;8(6):622-627.

55. Fan ST, Lo CM, Lai EC, et al. Perioperative nutritional support in patients undergoing hepatectomy for hepatocellular carcinoma. N Engl J Med. 1994;331(23):1547-1552.

56. Ichikawa K, Ichikawa K, Okabayashi T, et al. Oral supplementation of branched-chain amino acids reduces early recurrence after hepatic resection in patients with hepatocellular carcinoma: a prospective study. Surg Today. 2013;43(7):720-726.

57. Ishikawa Y, Ishikawa Y, Yoshida H, et al. Prospective randomized controlled study of short-term perioperative oral nutrition with branched chain amino acids in patients undergoing liver surgery. Hepatogastroenterology. 2010;57(99-100):583-590.

58. Kikuchi Y, Kikuchi Y, Hiroshima Y, et al. A Randomized Clinical Trial of Preoperative Administration of Branched-Chain Amino Acids to Prevent Postoperative Ascites in Patients with Liver Resection for Hepatocellular Carcinoma. Ann Surg Oncol. 2016;23(11):3727-3735.

59. Meng WC, Leung KL, Ho RL, Leung TW, Lau WY, Meng WC. Prospective randomized control study on the effect of branched-chain amino acids in patients with liver resection for hepatocellular carcinoma. Aust N Z J Surg. 1999;69(11):811-815.
60. Nagasue N. Long-term oral administration of branched chain amino acids after curative resection of hepatocellular carcinoma: A prospective randomized trial. Br J Surg. 1997;84(11):1525-1531.
61. Okabayashi T, Okabayashi T, Nishimori I, et al. Preoperative oral supplementation with carbohydrate and branched-chain amino acid-enriched nutrient improves insulin resistance in patients undergoing a hepatectomy: a randomized clinical trial using an artificial pancreas. Amino Acids. 2010;38(3):901-907.
62. Okabayashi T, Okabayashi T, Iyoki M, Sugimoto T, Kobayashi M, Hanazaki K. Oral supplementation with carbohydrate- and branched-chain amino acid-enriched nutrients improves postoperative quality of life in patients undergoing hepatic resection. Amino Acids. 2011;40(4):1213-1220.
63. Sun LC, Shih YL, Lu CY, et al. Randomized, controlled study of branched chain amino acid-enriched total parenteral nutrition in malnourished patients with gastrointestinal cancer undergoing surgery. Am Surg. 2008;74(3):237-242.
64. Togo S, Tanaka, Togo S, et al. Usefulness of granular BCAA after hepatectomy for liver cancer complicated with liver cirrhosis. Nutrition. 2005;21(4):480-486.
65. Wu Q, Wu Q, Zhang Y, Yang Y, Ge S, Xue Z. Intraoperative infusion of branched-chain amino acids in patients undergoing gastrointestinal tumor surgery. World J Surg Oncol. 2015;13:336.

Table S2.1 - Mortality (RCTs)

Study Author, Year	Tx Name	Sample size	No. deaths	Follow-up time	Notes
Fan, 1994	BCAAs-enriched PN	64	5	14.5 days	during surgical hospitalization
Fan, 1994	5% Dextrose + normal saline	60	9	16 days	during surgical hospitalization
Ichikawa, 2013	BCAAs	26	0	30 days	post-operative surgical mortality
Ichikawa, 2013	no added tx	30	0	30 days	post-operative surgical mortality
Ichikawa, 2013	BCAAs	26	1	1 yr	calculated from 1-yr survival rate
Ichikawa, 2013	no added tx	30	2	1 yr	calculated from 1-yr survival rate
Ichikawa, 2013	BCAAs	26	4	4 yrs	calculated from 4-yr survival rate
Ichikawa, 2013	no added tx	30	3	4 yrs	calculated from 4-yr survival rate
Ishikawa, 2010	BCAAs	11	0	7 days	
Ishikawa, 2010	no added tx	13	0	7 days	
Kikuchi, 2016	BCAAs	39	1	90 days	
Kikuchi, 2016	no added tx	38	0	90 days	
Meng, 1999	Aminoleban EN	21	4	30 days	post-operative mortality
Meng, 1999	no added tx	23	1	30 days	post-operative mortality
Meng, 1999	Aminoleban EN	21	7	512 days	mortality at last follow-up
Meng, 1999	no added tx	23	4	513 days	mortality at last follow-up

Nagasue, 1997	Aminoleban EN	67	34	35.8 mos	
Nagasue, 1997	no added tx	65	32	36 mos	
Okabayashi, 2010	Aminolevan EN	13	0	30 days	
Okabayashi, 2010	no added tx	13	0	30 days	
Okabayashi, 2011	Aminoleban EN	40	0	30 days	
Okabayashi, 2011	no added tx	36	0	30 days	
Sun, 2008	Amiparen 10% (enriched TPN, 30% BCAAs)	32	0	30 days	
Sun, 2008	Aminosyn 10% (standard TPN, 24% BCAAs)	32	1	30 days	

Table S2.2 - Recurrence (RCTs)

Study Author, Year	Tx Name	Sample size	No. recurrences	Follow-up time	Follow-up unit	Name of outcome
Ichikawa, 2013	BCAAs	26	7	30	months	cumulative HCC recurrence rate
Ichikawa, 2013	no added tx	30	17	30	months	cumulative HCC recurrence rate
Meng, 1999	Aminoleban EN	21	6	512	days	recurrence at last follow-up
Meng, 1999	no added tx	23	4	513	days	recurrence at last follow-up
Nagasue, 1997	Aminoleban EN	67	39	35.8	months	intrahepatic recurrence
Nagasue, 1997	no added tx	65	39	36	months	intrahepatic recurrence
Okabayashi, 2011	Aminoleban EN	48	3	NR	NR	recurrence
Okabayashi, 2011	no added tx	48	8	NR	NR	recurrence
Togo, 2005	BCAAs	21	2	12	months	tumour recurrence
Togo, 2005	no added tx	22	2	12	months	tumour recurrence

Table S2.3 - Liver Failure outcomes (RCTs)

Study Author, Year	Tx Name	Sample size	No. events*	Follow-up time	Follow-up unit	Name of outcome	Notes
Meng, 1999	Aminoleban EN	21	0	512	days	post-operative encephalopathy	p>0.05 non-significant
Meng, 1999	no added tx	23	0	513	days	post-operative encephalopathy	
Nagasue, 1997	Aminoleban EN	67	0%	12	months	% change from baseline in encephalopathy†	p=0.919 non-significant
Nagasue, 1997	no added tx	65	2.1%	12	months	% change from baseline in encephalopathy†	
Togo, 2005	BCAAs	19	0	12	months	post-operative encephalopathy	p>0.05 non-significant
Togo, 2005	no added tx	20	0	12	months	post-operative encephalopathy	
Meng, 1999	Aminoleban EN	21	0	512	days	post-operative flapping tremor	p>0.05 non-significant
Meng, 1999	no added tx	23	0	513	days	post-operative flapping tremor	
Nagasue, 1997	Aminoleban EN	67	0%	12	months	% change from baseline in flapping tremor	p=0.047 borderline significant
Nagasue, 1997	no added tx	65	2.1%	12	months	% change from baseline in flapping tremor	
Togo, 2005	BCAAs	19	0	12	months	post-operative flapping tremor	p>0.05 non-significant
Togo, 2005	no added tx	20	0	12	months	post-operative flapping tremor	

Kikuchi, 2016	BCAAs	39	3	NR	NR	post-operative liver failure‡	See Appendix Fig. S3.1
Kikuchi, 2016	no added tx	38	4	NR	NR	post-operative liver failure‡	
Meng, 1999	Aminoleban EN	21	3	512	days	post-operative liver failure	See Appendix Fig. S3.1
Meng, 1999	no added tx	23	1	513	days	post-operative liver failure	

*One study (Nagasue 1997) reported percent change instead of number of events.
†According to Sherlock's criteria.
‡Definition from Rahbari et al. in *Surgery*, 2011;149.

Table S2.4 - Ascites, Pleural Effusion and Edema (RCTs)

Study Author, Year	Tx Name	Sample size	No. events*	Follow-up time	Name of outcome
Fan, 1994	BCAAs-enriched PN	64	16	NR	post-operative ascites[†]
Fan, 1994	5% Dextrose + normal saline	60	30	NR	post-operative ascites[†]
Fan, 1994	BCAAs-enriched PN	64	9	NR	post-operative pleural effusion[‡]
Fan, 1994	5% Dextrose + normal saline	60	12	NR	post-operative pleural effusion[‡]
Ichikawa, 2013	BCAAs	26	2	NR	post-operative ascites
Ichikawa, 2013	no added tx	30	1	NR	post-operative ascites
Ishikawa, 2010	BCAAs	11	2	7 days	post-operative pleural effusion
Ishikawa, 2010	no added tx	13	1	7 days	post-operative pleural effusion
Kikuchi, 2016	BCAAs	39	12	7 days	ascites apparent on CT scan
Kikuchi, 2016	no added tx	38	13	7 days	ascites apparent on CT scan
Kikuchi, 2016	BCAAs	39	2	4 weeks	refractory post-operative ascites[§]
Kikuchi, 2016	no added tx	38	5	4 weeks	refractory post-operative ascites[§]
Kikuchi, 2016	BCAAs	39	18	7 days	pleural effusion apparent on CT scan
Kikuchi, 2016	no added tx	38	17	7 days	pleural effusion apparent on CT scan
Kikuchi, 2016	BCAAs	39	0	4 weeks	refractory post-operative pleural effusion
Kikuchi, 2016	no added tx	38	3	4 weeks	refractory post-operative pleural effusion

Meng, 1999	Aminoleban EN	21	8	512 days	persistent chest effusion ("major complication")
Meng, 1999	no added tx	23	10	513 days	persistent chest effusion ("major complication")
Meng, 1999	Aminoleban EN	21	2	512 days	chest effusion ("minor complication")
Meng, 1999	no added tx	23	3	513 days	chest effusion ("minor complication")
Nagasue, 1997	Aminoleban EN	67	0%	12 mos	% change from baseline in ascites
Nagasue, 1997	no added tx	65	2.0%	12 mos	% change from baseline in ascites (p=0.19 between groups, non-significant)
Nagasue, 1997	Aminoleban EN	67	0%	12 mos	% change from baseline in edema
Nagasue, 1997	no added tx	65	0%	12 mos	% change from baseline in edema
Togo, 2005	BCAAs	19	1[#]	12 mos	post-operative ascites[¶]
Togo, 2005	no added tx	20	1[#]	12 mos	post-operative ascites[¶]
Togo, 2005	BCAAs	19	1[#]	12 mos	post-operative edema
Togo, 2005	no added tx	20	0[#]	12 mos	post-operative edema

*One study (Nagasue 1997) reported percent change instead of number of events.

[†]Definition from study: Ascites was considered a complication when the accumulation was massive, leading to dyspnea or leakage through the abdominal wound, and a diuretic was needed for control.

[‡]Definition from study: Pleural effusion was considered a complication when the collection of fluid caused dyspnea and tapping was required for relief.

[§]Definition from study: Ascites of which daily output from an abdominal drain exceeded 10 mL/kg of preoperative body weight, or, after abdominal drain removal, ascites that failed to abate after therapeutic paracentesis, or recurred shortly afterward despite sodium restriction and diuretic treatment.

[¶]Definition from study: Ascites was evaluated by ultrasonography and physical examination.

[#]Estimated from Figure 2.

Table S2.5 - Hospital Length of Stay (RCTs)

Study Author, Year	Tx Name	Sample size	Hospital LOS (mean)	Variance type	Variance value (days)
Fan, 1994	BCAAs-enriched PN	64	14.5*	range	8-51
Fan, 1994	5% Dextrose + normal saline	60	16*	range	2-126
Ichikawa, 2013	BCAAs	26	13	SD	3
Ichikawa, 2013	no added tx	30	16	SD	7
Kikuchi, 2016	BCAAs	39	13.7	SD	11.2
Kikuchi, 2016	no added tx	38	12.7	SD	13.4
Meng, 1999	Aminoleban EN	21	10*	range	7-35
Meng, 1999	no added tx	23	16.5*	range	9-48
Wu, 2015	BCAAs	20	7.9	SD	1.8
Wu, 2015	AAs solution (4%)	21	8.1	SD	1.9
Wu, 2015	Normal saline	20	8.6	SD	2.4

*Only median value was reported.

Table S2.6 - Quality of Life (RCTs)

Study Author, Year	Tx Name	Sample size-Base-line	Mean value-Base-line	Var. type	Var. -B.	Sample size-Follow-up	Mean value-F.	Var. type	Var. -F.	Scale name (Units)	Time of F.	QOL name
Nagasue, 1997	Aminoleban EN	67	NA	NA	NA	67	10.3*	NR	NR	Karnov-sky scale (% change)	12 mos	Perfor-mance status ≤70
Nagasue, 1997	no added tx	65	NA	NA	NA	65	15.8	NR	NR	Karnov-sky scale (% change)	12 mos	Perfor-mance status ≤70
Okabaya-shi, 2011	Aminoleban EN	40	79†	NR	NR	40	91†	NR	NR	SF-36 (0-100)	12 mos	Bodily pain
Okabaya-shi, 2011	no added tx	36	82.5	NR	NR	36	83	NR	NR	SF-36 (0-100)	12 mos	Bodily pain
Okabaya-shi, 2011	Aminoleban EN	40	56	NR	NR	40	83	NR	NR	SF-36 (0-100)	12 mos	General health percep-tions

Okabaya-shi, 2011	no added tx	36	52	NR	NR	36	59	NR	NR	SF-36 (0-100)	12 mos	General health percep-tions
Okabaya-shi, 2011	Aminoleban EN	40	64	NR	NR	40	90	NR	NR	SF-36 (0-100)	12 mos	Mental health
Okabaya-shi, 2011	no added tx	36	69	NR	NR	36	73	NR	NR	SF-36 (0-100)	12 mos	Mental health
Okabaya-shi, 2011	Aminoleban EN	40	82.5	NR	NR	40	94	NR	NR	SF-36 (0-100)	12 mos	Physical function-ing
Okabaya-shi, 2011	no added tx	36	85	NR	NR	36	86	NR	NR	SF-36 (0-100)	12 mos	Physical function-ing
Okabaya-shi, 2011	Aminoleban EN	40	68	NR	NR	40	94	NR	NR	SF-36 (0-100)	12 mos	Role emotion-al
Okabaya-shi, 2011	no added tx	36	78	NR	NR	36	78	NR	NR	SF-36 (0-100)	12 mos	Role emotion-al
Okabaya-shi, 2011	Aminoleban EN	40	69	NR	NR	40	92	NR	NR	SF-36 (0-100)	12 mos	Role physical
Okabaya-shi, 2011	no added tx	36	75	NR	NR	36	78	NR	NR	SF-36 (0-100)	12 mos	Role physical
Okabaya-	Aminoleban EN	40	71	NR	NR	40	95	NR	NR	SF-36	12	Social

shi, 2011										(0-100)	mos	function-ing
Okabaya-shi, 2011	no added tx	36	75	NR	NR	36	81	NR	NR	SF-36 (0-100)	12 mos	Social function-ing
Okabaya-shi, 2011	Aminoleban EN	40	59	NR	NR	40	85	NR	NR	SF-36 (0-100)	12 mos	Vitality
Okabaya-shi, 2011	no added tx	36	58	NR	NR	36	62	NR	NR	SF-36 (0-100)	12 mos	Vitality

Notes:
*p=0.047 between groups.
†All absolute values in this study were estimated from Figure 3.

Table S2.7 - Anthropometrics (RCTs)

Study Author, Year	Tx Name	Sample size-Base-line	Value-Base-line (mean)	Var. type-B.	Var. B.	Sample size-Follow-up	Value-Follow-up (mean*)	Var. type-F.	Var. F.	Unit	Follow-up time	Name of outcome
Bonau, 1984	45% BCAAs-enriched low-leucine TPN + dextrose	6	NA	NA	NA	6	-1.5	SE	0.9	%	8 days	% change from baseline in body wt.
Bonau, 1984	25% BCAAs standard TPN (Aminosyn 7%) + dextrose	9	NA	NA	NA	9	-0.5 [p>0.05 between groups]	SE	0.4	%	8 days	% change from baseline in body wt.
Bonau, 1984	5% Dextrose in water	4	NA	NA	NA	4	NR	NR	NR	NR	NR	% change from baseline in body wt.

Fan, 1994	BCAAs-enriched PN	64	26	range	19-32	64	26	range	10 to 31	cm	15 days	midarm circum-ference (median)
Fan, 1994	5% Dextrose + normal saline	60	28	range	20-74	60	26	range	20 to 78	cm	15 days	midarm circum-ference (median)
Fan, 1994	BCAAs-enriched PN	64	NA	NA	NA	64	0	range	-10 to 6.5	kg	15 days	change from baseline in wt. (median)
Fan, 1994	5% Dextrose + normal saline	60	NA	NA	NA	60	-1.4 [p=0.01 between groups]	range	-7 to 1.7	kg	15 days	change from baseline in wt. (median)
Fan, 1994	BCAAs-enriched PN	64	10.6	range	3.2-21	64	11.2	range	3.7 to 20	mm	15 days	triceps skin-fold thick-ness (median)
Fan, 1994	5% Dextrose + normal saline	60	12	range	4-42	60	10.5	range	4 to 27	mm	15 days	triceps skin-fold

												thick-ness (median)
Ichikawa, 2013	BCAAs	26	28	SD	2.5	26	28.5†	SD	3	cm	6 mos	arm muscle circum-ference
Ichikawa, 2013	no added tx	30	27.5	SD	3	30	27.5†	SD	4	cm	6 mos	arm muscle circum-ference
Ichikawa, 2013	BCAAs	26	64	SD	9	26	66†	SD	10	kg	6 mos	body weight
Ichikawa, 2013	no added tx	30	61	SD	10	30	62†	SD	11	kg	6 mos	body weight
Nagasue, 1997	Aminoleban EN	67	NA	NA	NA	67	7‡	SD	8	%	12 mos	% change from baseline in body wt.
Nagasue, 1997	no added tx	65	NA	NA	NA	65	1‡ [p<0.01	SD	6	%	12 mos	% change from baseline

							between groups]					in body wt.
Okabaya-shi, 2011	Aminoleban EN	40	27	NR	NR	40	30.5§	NR	NR	cm	12 mos	arm muscle circum-ference
Okabaya-shi, 2011	no added tx	36	27	NR	NR	36	26.5§	NR	NR	cm	12 mos	arm muscle circum-ference
Okabaya-shi, 2011	Aminoleban EN	40	59	NR	NR	40	62§	NR	NR	kg	12 mos	body weight
Okabaya-shi, 2011	no added tx	36	58	NR	NR	36	56.5§	NR	NR	kg	12 mos	body weight
Sun, 2008	Amiparen 10% (enriched TPN, 30% BCAAs)	32	21.3	SD	3.5	32	21.1	SD	3.7	kg/m^2	7 days	body mass index
Sun, 2008	Aminosyn 10% (standard TPN, 24% BCAAs)	32	20.7	SD	4.1	32	19.1	SD	4.1	kg/m^2	7 days	body mass index
Sun, 2008	Amiparen 10% (enriched TPN, 30% BCAAs)	32	53.8	SD	10	32	53.2	SD	10.1	kg	7 days	body weight
Sun, 2008	Aminosyn 10% (standard TPN,	32	54.1	SD	11.6	32	53	SD	9.1	kg	7 days	body weight

	24% BCAAs)											

*When only a median value was reported, this is noted in the last column.
†Estimated from Figure 3.
‡estimated from Figure 1.
§estimated from Figure 4.

Table S2.8 - Immune Cells and Inflammatory Markers (RCTs)

Study Author, Year	Tx Name	Sample size-Base-line	Value-Base-line	Var.-B.	Sample size-Follow-up	Value-Follow-up	Var.-F.	Unit	Follow-up time	Name of outcome
Fan, 1994	BCAAs-enriched PN	64	320	range 144-680	64	331*	range 156-792	mg/dl	15 days	serum IgA (median)
Fan, 1994	5% Dextrose + normal saline	60	280	range 79-794	60	287*	range 142-564	mg/dl	15 days	serum IgA (median)
Fan, 1994	BCAAs-enriched PN	64	1500	range 926-2850	64	1415†	range 866-3060	mg/dl	15 days	serum IgG (median)
Fan, 1994	5% Dextrose + normal saline	60	1378	range 682-2708	60	1259†	range 812-2414	mg/dl	15 days	serum IgG (median)
Fan, 1994	BCAAs-enriched PN	64	134	range 36-390	64	162‡	range 28-446	mg/dl	15 days	serum IgM (median)
Fan, 1994	5% Dextrose + normal saline	60	124	range 47-352	60	145‡	range 54-386	mg/dl	15 days	serum IgM (median)
Sun, 2008	Amiparen 10% (enriched TPN, 30% BCAAs)	32	83	SD 30.1	32	60.3	SD 29.2	μg/mL	7 days	C-reactive protein (mean)
Sun, 2008	Aminosyn 10% (standard TPN,	32	78.7	SD 25.1	32	90.2	SD 30.3	μg/mL	7 days	C-reactive protein

	24% BCAAs)									(mean)
Sun, 2008	Amiparen 10% (enriched TPN, 30% BCAAs)	32	10,135	SD 3,450	32	8,329	SD 3,473	µL	7 days	WBC count (mean)
Sun, 2008	Aminosyn 10% (standard TPN, 24% BCAAs)	32	8,350	SD 3,984	32	9,117	SD 4,106	µL	7 days	WBC count (mean)

*Normal IgA reference range cited in the study: 66-344 (according to Ward and Randal, 1983: American College of Surgeons).
†Normal IgG reference range cited in the study: 540-1663 (according to Ward and Randal, 1983: American College of Surgeons).
‡Normal IgM reference range cited in the study: 39-329 (according to Ward and Randal, 1983: American College of Surgeons).

Table S2.9 - Ammonia Level (RCTs)

Study Author, Year	Tx Name	Sample size-Baseline	Mean value-Baseline	Var. type	Var. value-Baseline	Sample size-Follow-up	Mean value-Follow-up	Var. type	Var. value-Follow-up	Unit	Follow-up time
Ishikawa, 2010	BCAAs	11	42	NR	NR	11	36*	NR	NR	mcg/ dL	7 days
Ishikawa, 2010	no added tx	13	45	NR	NR	13	42*	NR	NR	mcg/ dL	7 days
Nagasue, 1997	Aminoleban EN	67	56	SD	28	67	51	SD	24	mcg/ dL†	12 months
Nagasue, 1997	no added tx	65	57	SD	26	65	48	SD	26	mcg/ dL†	12 months

Notes:
*Estimated from Figure 1.
†Possible units error in the publication since was reported as "g/dl".

Table S2.10 - Prothrombin Time (RCTs)

Study Author, Year	Tx Name	Sample size-B.	Mean value-Base-line	Var. type	Var.-B.	Sample size-F.	Mean value-Follow-up	Var. type	Var.-F.	Unit	Follow-up time
Kikuchi, 2016	BCAAs	39	1.071	SD	0.076	39	1.185	SD	0.216	NR	14 days
Kikuchi, 2016	no added tx	38	1.073	SD	0.167	38	1.144	SD	0.141	NR	14 days

Table S2.11 - Blood Loss (RCTs)

Author, Year	Tx Name	Sample size	Blood loss (mean*)	Variance type	Variance value	Unit	Name of outcome
Kikuchi, 2016	no added tx	38	578.5	SD	492.1	ml	Intra-operative bleeding
Kikuchi, 2016	BCAAs	39	665.7	SD	528.9	ml	Intra-operative bleeding

Ichikawa, 2013	Livact (BCAA)	26	492	SD	329	ml	Estimated blood loss
Ichikawa, 2013	no added tx	30	716	SD	704	ml	Estimated blood loss
Okabayashi, 2011	Aminoleban EN	40	945	SD	827	ml	Estimated operative blood loss
Okabayashi, 2011	no added tx	36	676	SD	695	ml	Estimated operative blood loss
Ishikawa, 2010	BCAAs	11	802.7	SD	350.6	ml	Total operative blood loss
Ishikawa, 2010	no added tx	13	838.8	SD	597.6	ml	Total operative blood loss
Okabayashi, 2010	Aminolevan EN	13	1252	SD	1205	ml	Estimated operative blood loss
Okabayashi, 2010	Control	13	669	SD	575	ml	Estimated operative blood loss
Fan, 1994	BCAAs + Dextrose + Lipid emulsion + Vitamins + trace minerals	64	2.6	range	0.3-20	Liters	Intra-operative blood loss (median)
Fan, 1994	Normal diet + 5% Dextrose + Normal saline	60	1.9	range	0.4-10.5	Liters	Intra-operative blood loss (median)

*When only a median value was reported, this is noted in the last column.

APPENDIX 3. Additional RCT Forest Plots

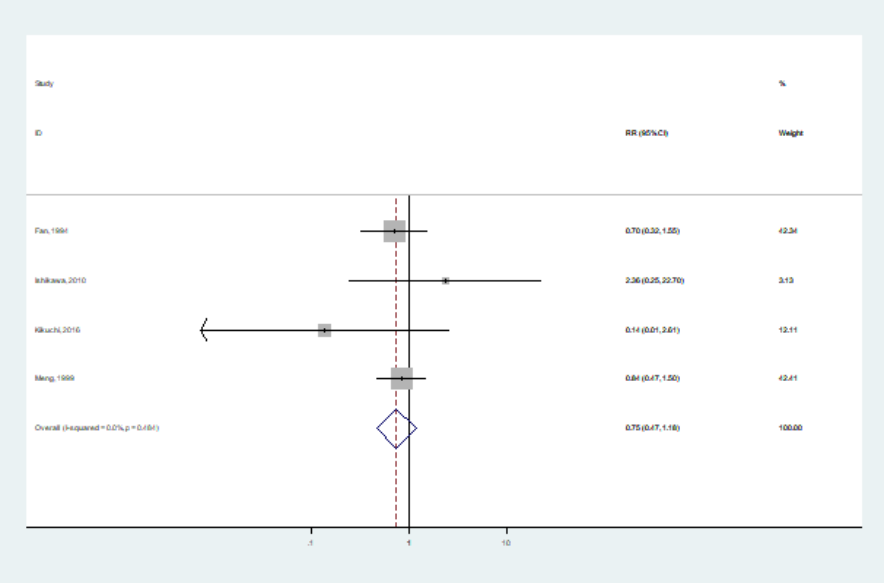


Figure S3.1. Forest Plot of RCTs Meta-Analysis on Pleural Effusion

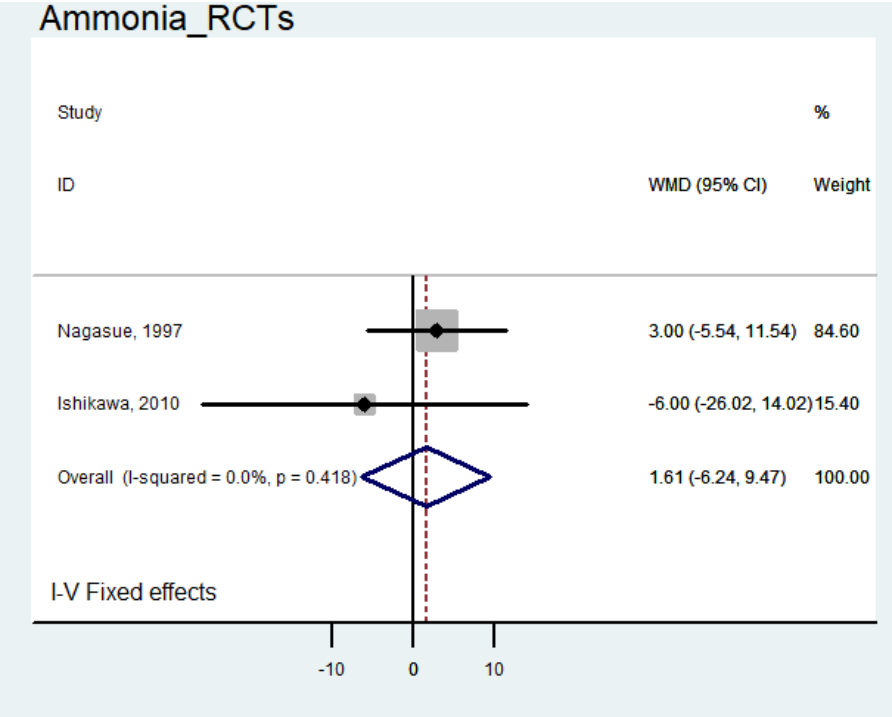


Figure S3.2. Forest Plot of RCTs Meta-Analysis on Ammonia Level

APPENDIX 4. Tables of Results from Observational Studies and Alphabetical List of Included Observational Studies

Note: The number below corresponds to the reference number cited in the main report's References list:

66. Ardito F, Lai Q, Rinninella E, et al. The impact of personalized nutritional support on postoperative outcome within the enhanced recovery after surgery (ERAS) program for liver resections: results from the NutriCatt protocol. Updates Surg. Sep 2020;72(3):681-691. doi:10.1007/s13304-020-00787-6

67. Huang HH, Wu PC, Kang SP, et al. Postoperative hypocaloric peripheral parenteral nutrition with branched-chain-enriched amino acids provides no better clinical advantage than fluid management in nonmalnourished colorectal cancer patients. Nutr Cancer. 2014;66(8):1269-1278.

68. Nakajima H, Yokoyama Y, Inoue T, et al. Clinical Benefit of Preoperative Exercise and Nutritional Therapy for Patients Undergoing Hepato-Pancreato-Biliary Surgeries for Malignancy. Ann Surg Oncol. Jan 2019;26(1):264-272. doi:10.1245/s10434-018-6943-2

69. Nanno Y, Toyama H, Terai S, et al. Preoperative Oral Branched-Chain Amino Acid Supplementation Suppresses Intraoperative and Postoperative Blood Lactate Levels in Patients Undergoing Major Hepatectomy. JPEN J Parenter Enteral Nutr. Feb 2019;43(2):220-225. doi:10.1002/jpen.1445

70. Okabayashi T, Okabayashi T, Nishimori I, et al. Effects of branched-chain amino acids-enriched nutrient support for patients undergoing liver resection for hepatocellular carcinoma. J Gastroenterol Hepatol. 2008;23(12):1869-1873.

71. Okabayashi T, Okabayashi T, Nishimori I, et al. The benefit of the supplementation of perioperative branched-chain amino acids in patients with surgical management for hepatocellular carcinoma: a preliminary study. Dig Dis Sci. 2008;53(1):204-209.
72. Shirabe K, Shirabe K, Yoshimatsu M, et al. Beneficial effects of supplementation with branched-chain amino acids on postoperative bacteremia in living donor liver transplant recipients. Liver Transpl. 2011;17(9):1073-1080.

Table S4.1 - Mortality (Observational)

Study Author, Year	Tx Name	Sample size	No. Deaths	Follow-up time	Follow-up unit	MORTALITY
Ardito, 2020	ERAS + BCAAs + personalized diet	113	0	90	days	postoperative mortality
Ardito, 2020	ERAS program	261	1	90	days	postoperative mortality
Huang, 2014	high-dose BCAAs + fat	21	0	7	days	in-hospital mortality
Huang, 2014	low-dose BCAAs + fat	20	0	7	days	in-hospital mortality
Huang, 2014	traditional fluid management	20	0	7	days	in-hospital mortality
Nakajima, 2019	Exercise + BCAAs + Vitamin D	76	0*	90	days	postoperative mortality
Nakajima, 2019	no prehabilitation	76	0*	90	days	postoperative mortality
Nanno, 2019	AminoValue	146	0	30	days	postoperative mortality
Nanno, 2019	no added tx	129	0	30	days	postoperative mortality
Okabayashi, 2008a	Aminoleban EN	40	3	12	months	1-year mortality rate
Okabayashi, 2008a	no added tx	72	11	12	months	1-year mortality rate
Okabayashi, 2008a	Aminoleban EN	40	9	36	months	3-year mortality rate
Okabayashi, 2008a	no added tx	72	13	36	months	3-year mortality rate
Shirabe, 2011	BCAAs	129	8	NR	NR	in-hospital mortality†
Shirabe, 2011	no added tx	107	10	NR	NR	in-hospital mortality†

*Propensity score matched.
†Calculated from number in hospital.

Table S4.2 – Liver Failure (Observational)

Study Author, Year	Tx Name	Sample size	No. Liver Failure	Follow-up time	LIVER FAILURE	p-value
Nakajima, 2019	Exercise + BCAAs + Vitamin D	76	4	NR	Liver failure grade ≥B*	0.367
Nakajima, 2019	no prehabilitation	76	1	NR	Liver failure grade ≥B*	

*Propensity score matched. Study definition of the International Study Group of Liver Surgery (Rahbari 2011; Koch 2011).

Table S4.3 – Post-operative Infections (Observational)

Study Author, Year	Tx Name	Sample size	No. Infections	Follow-up time	INFECTIONS
Nakajima, 2019	Exercise + BCAAs + Vitamin D	76	5*	NR	Bacteremia
Nakajima, 2019	no prehabilitation	76	6*	NR	Bacteremia
Nakajima, 2019	Exercise + BCAAs + Vitamin D	76	8*	NR	Cholangitis
Nakajima, 2019	no prehabilitation	76	8*	NR	Cholangitis
Nakajima, 2019	Exercise + BCAAs + Vitamin D	76	19*	NR	Infectious complications
Nakajima, 2019	no prehabilitation	76	25*	NR	Infectious complications
Nakajima, 2019	Exercise + BCAAs + Vitamin D	76	11*	NR	Intraabdominal abscess
Nakajima, 2019	no prehabilitation	76	14*	NR	Intraabdominal abscess
Nakajima, 2019	Exercise + BCAAs + Vitamin D	76	1*	NR	Pneumonia
Nakajima, 2019	no prehabilitation	76	1*	NR	Pneumonia
Nakajima, 2019	Exercise + BCAAs + Vitamin D	76	2*	NR	Surgical-site infection
Nakajima, 2019	no prehabilitation	76	3*	NR	Surgical-site infection
Okabayashi, 2008a	Aminoleban EN	40	2	NR	Surgical site infection
Okabayashi, 2008a	no added tx	72	11	NR	Surgical site infection
Shirabe, 2011	BCAAs	129	11	90 days	Bacteremia†
Shirabe, 2011	no added tx	107	17	90 days	Bacteremia†

*Propensity score matched.
†Study defined as "isolation of bacteria (other than common skin contaminants) from a single blood culture obtained within 90 days of transplantation or as the presence of clinical symptoms or signs of infection".

Table S4.4 - Ascites (Observational)

Study Author, Year	Tx Name	Sample size	No. Ascites	Follow-up time	ASCITES
Okabayashi, 2008a	Aminoleban EN	40	3	NR	post-operative ascites
Okabayashi, 2008a	no added tx	72	12	NR	post-operative ascites

Table S4.5 – Hospital Length of Stay (Observational)

Study Author, Year	Tx Name	Sample size	Hospitalization length (days)	Variance type	Variance value	HOSPITAL LENGTH OF STAY
Ardito, 2020	ERAS + BCAAs + personalized diet	113	6*	IQR	5-8	hospital stay (median)
Ardito, 2020	ERAS program	261	7*	IQR	5-10	hospital stay (median)
Huang, 2014	high-dose BCAAs + fat	21	12.5	SE	0.7	postop hospital stay (mean)
Huang, 2014	low-dose BCAAs + fat	20	12.9	SE	1.1	postop hospital stay (mean)
Huang, 2014	traditional fluid management	20	12.5	SE	0.7	postop hospital stay (mean)
Nakajima, 2019	Exercise + BCAAs + Vitamin D	76	23†	IQR	16–34	postop hospital stay (median)
Nakajima, 2019	no prehabilitation	76	30†	IQR	21–40	postop hospital stay (median)
Okabayashi, 2008a	Aminoleban EN	40	15.5	SD	4.9	hospital stay (mean)
Okabayashi, 2008a	no added tx	72	22.3	SD	15.2	hospital stay (mean)

*Propensity score matching (113 patients per group) for long LOS >8 days found OR= 0.91 (95% CI, 0.86 to 0.97; p= 0.002).
†Propensity score matched.

Table S4.6 – Anthropometrics (Observational)

Study Author, Year	Tx Name	Sample size-Baseline	BMI - Baseline (kg/m^2)	Variance value-Baseline (SE)	Sample size-Follow-up	BMI - Follow-up (kg/m^2)	Variance value-Follow-up (SE)	Follow-up time
Huang, 2014	high-dose BCAAs + fat	21	24.1	0.8	21	23.4	0.8	7 days
Huang, 2014	low-dose BCAAs + fat	20	22.7	0.6	20	21.6	0.6	7 days
Huang, 2014	traditional fluid management	20	24.9	0.6	20	23.5	0.6	7 days

Table S4.7 – Immune Cells (Observational)

Study Author, Year	Tx Name	Sample size-Baseline	Mean value-Baseline (cells/ mm^3)	Var. value-Baseline	Sample size-Follow-up	Mean value-Follow-up (cells/ mm^3)	Var. value-Follow-up	Follow-up time	LYMPHO-CYTES	p-value
Huang, 2014	high-dose BCAAs + fat	21	1462	SE: 128	21	1157	SE: 91	7 days	total lymphocyte count	0.163*
Huang, 2014	low-dose BCAAs + fat	20	1649	SE: 146	20	1232	SE: 85	7 days	total lymphocyte count	0.223*
Huang, 2014	traditional fluid management	20	1686	SE: 124	20	1426	SE: 87	7 days	total lymphocyte count	NA
Okabaya-shi, 2008b	Aminoleban EN	13	2000	NR	13	2000†	NR	6 months	peripheral lymphocyte count	>0.05
Okabaya-shi, 2008b	no added tx	28	1500	NR	28	1500†	NR	6 months	peripheral lymphocyte count	

*Compared to control, the change from baseline after adjusting for pre-operative BCAA, protein, and calorie intake, BMI and nitrogen balance.
†Estimated from Figure 1.

Table S4.8 – Blood Loss (Observational)

Study Author, Year	Tx Name	Sample size	Blood Loss (ml)	Variance type	Variance value	BLOOD LOSS
Nakajima, 2019	Exercise + BCAAs + Vitamin D	76	837[†]	IQR	596–1384	Intra-operative blood loss (median)
Nakajima, 2019	no prehabilitation	76	936[†]	IQR	645–1426	Intra-operative blood loss (median)
Nanno, 2019	AminoValue	146	772	SE	54	operative blood loss (mean)
Nanno, 2019	no added tx	129	1340	SE	118	operative blood loss (mean)
Okabayashi, 2008a	Aminoleban EN	40	516	SD	354	operative blood loss (mean)
Okabayashi, 2008a	no added tx	72	821	SD	552	operative blood loss (mean)
Shirabe, 2011	BCAAs	129	7388	SD	9031	Intra-operative blood loss (mean)
Shirabe, 2011	no added tx	107	6448	SD	6547	Intra-operative blood loss (mean)

[†]Propensity score matched.